Aalborg Universitet

AALBORG UNIVERSITY

What counts - Thinkering with Visual Mapping Toward Relational More than Human Temporality in Participatory Design

Zohar, Hadas

Published in:
PDC '26

DOI (link to publication from Publisher):
10.1145/3796624.3796635

Creative Commons License
CC BY 4.0

Publication date:
2026

Document Version
Publisher's PDF, also known as Version of record

Link to publication from Aalborg University

Citation for published version (APA):
Zohar, H. (2026). What counts - Thinkering with Visual Mapping Toward Relational More than Human Temporality in Participatory Design. In D. Selloni, A. Meroni, L. Galluzzo, D. Fassi, D. de Sainz Molestina, S. Broadbent, A. D. Campbell, L. Forlano, E. Giaccardi, F. Jegou, L. Kimbell, S. Maffei, I. Mariani, M. Mortati, M. Pasari, J. de la Rosa, S. Roudavski, R. Rugeles, S. Schade, P. Scupelli, M. Tironi, O. Tomico, F. Vergani, B. Villari, J. S. Baek, ... T. Franqueira (Eds.), *PDC '26: Full Papers, Peace, Dialogue, Coexistence. Designing for living together* (Vol. Vol. 1: Full Papers, pp. 465-478). Association for Computing Machinery (ACM). https://doi.org/10.1145/3796624.3796635

General rights
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

- Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying the publication in the public portal -

Take down policy
If you believe that this document breaches copyright please contact us at vbn@aub.aau.dk providing details, and we will remove access to the work immediately and investigate your claim.

Downloaded from vbn.aau.dk on: 11 07, 2026

What counts - Thinkering with Visual Mapping Toward Relational More than Human Temporality in Participatory Design

Hadas Zohar*
Architecture, Design and Media Technology
Aalborg University
Copenhagen, Denmark
hzo@create.aau.dk

Abstract

This paper proposes a relational approach and practical tools for understanding more-than-human temporality in participatory design, countering perceptions of time as objective and individual. To address this, I revisit studies on relational temporality across various fields and ask: How can participatory design support a relational understanding of more-than-human time? The findings demonstrate how tinkering with participatory visual mapping can benefit a relational understanding of temporality in participatory design, encompassing relations between different time scales, multi-species temporal dependencies, the human capacity to perceive and synchronise with more-than-human time, and the creation of shared time. I then suggest that visual mapping can help us develop our 'relational temporal sensitivity' by recognising our limitations in perceiving beyond human richness and diversity, and by expanding our perception of more-than-human relational time. The study is grounded in three participatory design workshops conducted in Denmark and Sweden in 2024.

CCS Concepts

• **Human-centered computing**; • **Interaction design**; • **Interaction design theory, concepts and paradigms**; • **Visualization**; • **Visualization systems and tools**;

Keywords

Temporality, participatory design, visual mapping, relationality, more than human

ACM Reference Format:
Hadas Zohar. 2026. What counts - Thinkering with Visual Mapping Toward Relational More than Human Temporality in Participatory Design. In *Participatory Design Conference 2026, Vol. 1: Full Papers (PDC 2026 Vol. 1), June 15–19, 2026, Milan, Italy*. ACM, New York, NY, USA, 14 pages. https://doi.org/10.1145/3796624.3796635

1 Introduction

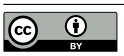
Scholars across various disciplines argue that our perception of time is not an innate feature of the mind but emerges from interactions with technical, material, embodied, and contextual practices [7, 22, 29, 42, 54]. Similarly, time in design is not a neutral backdrop but a relational phenomenon that accommodates multiple temporalities. Within design processes, formal structures such as schedules, calendars, and milestones regulate collaboration and establish shared rhythms. Alongside these institutionalised structures, human actors enact their own subjective temporalities—driven by project priorities, individual routines, and situated rhythms of work [6, 23, 50, 64]. When these dimensions converge in practice, they often privilege linear, human-centred notions of progress that accelerate production and delivery [26, 27, 40], while marginalising slower, cyclical, or ecological timescales [13, 34, 57]. While participatory design has long addressed time as a social phenomenon [23, 36, 37, 43, 68], its temporal lens remains anthropocentric mainly. This paper extends the discussion toward ecological and more-than-human temporalities.

Emerging pluriversal approaches within design have been challenging the universalist logics embedded in dominant design paradigms. Rather than assuming a single, linear temporality, pluriversal design embraces the notion that "things are their relations" [26:87] and foregrounds interdependence and entanglement as foundational principles for understanding complexity and enacting meaningful change [2, 45]. It underscores the idea that humans are fundamentally relational beings entangled with diverse entities and systems [26]. While the relational dimensions of temporality have long been explored in the social sciences and in the contexts of more-than-human time in biology and ecology [56], design research is only beginning to explore their intersection [36]. As participatory design increasingly integrates relational thinking [35] in areas such as human collaboration [68], workshop facilitation [32], and more-than-human participation [3, 46, 62], the temporal dimensions of these complex interactions remain underexplored.

Importantly, attention to plural temporalities is not a new concept. Historically, indigenous and pre-industrial societies developed strong attunements to ancestral temporality through practices of ritual, storytelling, and memory [14], as well as to the rhythms of the land and non-human world – astronomy, weather systems, flora and fauna – through long-term tracking and observation [1, 11, 33]. Thus, they have developed nuanced understandings of time as circular, layered, and non-linear. Yet, from the launching of GMT (Greenwich Mean Time) in 1884 onwards, temporality was increasingly reduced to a global universal system, imposing the misleading idea that time is singular, standardised, and unified [11, 28, 52, 59, 69].

Re-engaging with the relational dimensions of more-than-human time is vital for contemporary design. First, temporality is an essential aspect of diversity [25, 73]. Actors have their own peculiar ways of experiencing time, so multiple, heterogeneous times intersect,

*Correspondence author



This work is licensed under a Creative Commons Attribution 4.0 International License.
PDC 2026 Vol. 1, Milan, Italy
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2105-2/2026/06
https://doi.org/10.1145/3796624.3796635

interfere with, and intermingle [10]. As Fry [28] mentioned, no one has any experiential knowledge of what has been constructed as universal time. Therefore, understanding what time means helps illuminate the subjectivities, motivations, and interdependencies of diverse actors [8, 10, 53, 65]. Ignoring the dimensionality of temporality risks oversimplification, erases nuanced temporal dynamics, and suppresses complexity [21, 69]. Second, our tendency to think in temporal scales that match our lifetimes [28, 71] counters the human impact beyond our lifetimes, affecting geological timeframes and positioning humans as a geologic force. These temporal tensions, which strongly affect all beings on Earth, call us to recalibrate our temporal attention [10], access a plural understanding of time [41], and realise how our temporality is entangled, affects others, and is affected by them [19].

In this paper, I revisit studies on relational temporality from various fields and ask: How can participatory design support a relational understanding of time that extends beyond the human? Drawing on insights from three workshops, I explore how participatory visual mapping can engage with temporal complexity, including connecting disparate time scales, highlighting multispecies temporal dependencies, the human capacity to perceive and synchronise with more-than-human time, and creating shared time. I then suggest that participatory visual mapping can help us recognise our limitations in perceiving the richness and diversity of more-than-human time, develop relational temporal sensitivity, and expand our perception of more-than-human relational time.

The structure of this paper is as follows: In Section Two, I review existing literature according to the categories 'relational approaches to temporality', 'participatory design work that embraces a relational approach to temporality' and 'participatory visual mapping'. Section Three explains the methodology used. Section Four presents the main findings of the workshops, and Section Five contains the discussion and articulates the paper's main contribution.

2 Literature review

2.1 Relational approaches to temporality

Scholars from the social sciences and humanities, as well as design researchers, have proposed various perspectives to understand the relational quality of time. The choice to draw from these fields reflects that until recently, human-centred aspects of temporality were often studied separately from the 'time of nature' [9]. We can roughly divide the work on the relational dimensions of temporality into four categories: temporality resulting from relations and collaborations among humans; human temporalities that affect non-human temporalities; non-human temporalities that affect human temporalities; and the convergence of multiple-species temporalities.

Within the context of time as a result of relations and collaborations among humans, Bastian [7] perceives temporality as a set of conventions resulting from a shared agreement within communities on what to prioritise and what is significant to them (e.g., working during the day or the night) and who to consider as part of this coordination. In design research, Pschetz & Bastian [60:169] proposed the "temporal design" frame, centred on social coordination. They suggest leveraging social time "as emerging out of relations between cultural, social, economic and political forces" to allow a more specific focus on issues of ethics, equality, power, and social management. They identify it as a potential for designers to look beyond a single context, to ask how to promote interconnectedness, and to prioritise the relationships and interactions that influence an individual or group's temporal experiences.

Other writers adopt an ecological angle, emphasising the relationships that arise when human temporalities affect non-human temporalities. Pschetz et al. [57] gathered multiple examples from the field of ecology of how the temporalities of urbanisation and industrialisation impact species. Ingold [37:152] suggests considering the actions of social life in the landscape through the concept of "dwelling perspective." He proposed a view of the landscape as accumulated human temporalities, functioning as a "record of - and testimony to - the lives and works of past generations who have dwelt within it, and in so doing, have left there something of themselves." Here, the landscape serves as a mediator of human relations over time. According to Bjornerud [13], geologists can interpret the landscape in terms of its parallel histories and timelines, which she links to a "polytemporal" way of thinking. Another way human temporalities can affect non-human temporalities is through care. Puig de la Bellacasa [58:171] writes about the 'temporality of care' that she explores concerning soil and suggests how "care time" entails "making time" to get involved with "a diversity of timelines." She identifies the impact of care time as suspending the future and descending into the present, "thickening it with myriad multilateral demands." Bessai et al. [11] adopted a similar approach in designing a festival stage intended for use by both humans and other species, arguing that tangible engagement makes the complexity of multiple temporalities easier to perceive.

The view of how non-human temporalities affect human temporalities is explored in the work of Joks and Law [38:5], who researched Sámi fishing practices, applying a flow-based temporality rather than the Western-colonial clock used by fisheries biologists in Norway. The temporality assigned to Sámi fishing practices prioritises movement that responds to the relationships between tasks and the weather, and as such, cannot be predicted and is entirely dependent on other-than-human temporality. Pschez et al. [57] collected various cases from biology, ecology, and medical science research, demonstrating how non-human organisms affect human bodies, emotions, and inclinations toward specific activities.

The relational temporality created by the coming together of multiple species is evident in Latour's thinking [42], which argues against the binary division of 'objective' and 'subjective' time. Our experiences of time, he writes, are a consequence of how bodies relate to one another. For Latour, time is not a priori given but generated through relationships, the nature of the 'others' encountered, and the complexity they entail. Time is determined by the degree of obedience of the interconnected actors and their positionality as intermediaries or mediators, which may or may not facilitate a smooth interaction. Tsing [70] develops her thinking on more-than-human relationality through the concepts of entanglement and encounter. She rejects the familiar perspective of analysing a single creature or relationship at a time, including humans. Instead, she suggests the notion of polyphonic assemblages as "open-ended gathering" (p. 218) and "performances of livability" (p. 157). The temporal dimension appears in her work through the metaphor of music. Tsing unpacks the relational qualities between species

through the notion of different melodies coming together in moments of harmony and dissonance, as an alternative to "the unified progress of time" (p. 34) and the "puls of business" that "moves us forward" (p. 132). Another temporal consideration Tsing brings up is deciding what counts as a disturbance. In other words, what interrupts the flow? In her view, there is no standard; "disturbance matters in relation to how we live" (p. 161). Gan [29] uses rice cultivation to explore how relational capacities are shaped by time, arguing that matter exists only through relations, relations unfold through duration, and duration emerges in difference. Varieties thus arise from multispecies temporal coordination. She examines temporality through three lenses: long historical trajectories, recursive cycles of attunement, and moments of encounter. In contrast, Rifkin [61] rejects the notion of time as a shared container, instead asking how beings influence one another without occupying the same timeline. Abram [1] adds that encounters with other species require abandoning human historical time in favour of the land's cyclical rhythms. Similarly, Bastian and Byliss Hawitt [9] propose phenology—the study of seasonal life cycles—to understand cross-species temporal relations.

One question that appears throughout the various categories of relational temporality is "Who or what counts" [7, 11, 42, 44, 49]; which actors does design bring together, who, or what, is prioritised [11], and what are the emergent engagement that must be realised, which are dependent upon the physical affordance of the environment [49]. Relational temporality is defined by the deliberate decision of who we choose to include and what we value.

2.2 Participatory design work that embraces a relational approach to temporality

The temporal dimension is implicit in design, as design is about devising a change of what is not yet there [67] and, therefore, "has a direction into the future" [43:471]. Wieczorek and Forlano [72] described time as "the consistent, silent beating heart of design work". Although time is implicit in many design methods, such as user journeys, diary studies, temporal probes, and storytelling [74], within participatory design, there are only a few examples of how design methods explore and leverage relational temporality. This section will mention some of the few works that addressed the topic.

The multiple temporalities that come together in participatory processes were analysed by Del Gaudio et al. [20], who noted that temporal misfits may be a common occurrence in participatory design, and by Saad-Sulonen et al. [64], who defined five temporal lenses for participation. Armstrong et al. [4] argued for the importance of time in building trust among partners and collaborators in participatory research, Saad-Sulonen & Watkin [65] articulated the various temporalities coming together in design and research time, Hayes et al. [34] explored the temporal flow of participation by looking at the participation, project, and process lines, exploring the temporalities of every storyline, including its rhythms, tempos, urgencies and timings, and Oogjes et al. [54] developed a temporal vocabulary to cultivate a stronger temporal sensitivity in research-through-design.

Participatory design work that used design methods to better understand multiple temporal dimensions, include mock-ups for alternative narratives and co-design technological prototypes, exploring how engaging with different kinds of pasts can inform future perspectives and lead to new narratives of the past emerging [39], co-create, spatiotemporal mapping to redefine the design space by situating it in a historical context within heritage development [75]. These investigations are mainly centred around a historical notion of time.

Several scholars have explored relational approaches to more-than-human time. Pschetz et al. [57] conducted an interdisciplinary workshop that brought together designers, biologists, and environmental humanities researchers, aiming to generate multiple temporalities and (a)synchronise visible, catered towards developing practical design concepts to support critical discussions on the effects of climate and biodiversity changes. Oktay et al. [53] explore the temporalities of more-than-human care, including past, present, and future chronologies that may be enacted and lived concurrently in care partnerships. They point out how care practice forces us to abandon narratives of progress and an accelerated sense of the future in favour of respecting the slow pace and rhythm of more-than-human time. Masui and Kosaka [50] argued for the potentiality of embracing temporal discrepancies by examining the historical development of Katsuobushi. Zohar et al. [56] investigated how visual mapping can assist stakeholders engaged in participatory design processes in thinking through various aspects of more-than-human time.

Even though participatory design has long treated time as a social and relational concern, focusing on how temporal categories are identified and negotiated through human participation, collaboration, and shared rhythms [23, 36, 37, 43, 68] —it seldom questions the boundaries of these negotiations: which temporalities are included or excluded, and what kinds of worlds and relations these exclusions sustain. This reveals an anthropocentric framing, in which time is primarily understood through human coordination rather than as a quality that shapes and is shaped by nonhuman and ecological processes. Addressing this limitation calls for conceiving time not only as a social construct but also as an ecological and ethical matter, recognising how diverse temporalities coexist, interfere, and co-constitute one another within shared worlds. From this standpoint, expanding the temporal lens of participatory design becomes essential for engaging with more-than-human relations and reorienting design toward ecological interdependence and responsibility [6, 7, 30].

2.3 Participatory visual mapping

Although mapping is widely used in design, the concept of visual mapping remains loosely defined within participatory design literature [74]. References to the term typically point to different visualisation techniques [18, 30, 71, 74, 75], but in participatory design, visual maps are more often understood as processual tools [41]. They support shared reflection, helping stakeholders to explore relationships and make sense of complex conditions. Rather than static representations, they evolve throughout the design process [24, 41], opening space for subjectivity and reflexivity, and fostering dialogue, sensory engagement, and collective exploration. In this sense, visual mapping can also be seen as an aesthetic experiment that surfaces what is unconscious or taken for granted, while

activating alternative ways of experiencing reality—intellectually, affectively, and relationally [48].

To conclude, participatory design has increasingly adopted relational perspectives, foregrounding interdependencies among people, places, and more-than-human actors. Yet, despite this growing attention to relationality, less consideration has been given to how temporality itself is experienced, structured, and enacted within these entanglements [29]. The concept of "relational sensitivity" [32, 46] suggests that designers can reflect on their positionality and the ethical implications of their relationships with others. Extending this idea, I introduce the notion of temporal relational sensitivity—an orientation that foregrounds how time itself is sensed, co-created, and negotiated across species and material systems. Cultivating such sensitivity is essential for participatory practices that seek to engage with complexity and move beyond anthropocentric models of time. The following sections explore how participatory visual mapping develops and operationalises this concept, asking: How can participatory design advance a relational understanding of more-than-human time?

3 THE METHODOLOGY

3.1 The methodological framework

This paper adopts a research-through-design methodology, emphasising an iterative process composed of three interconnected components. The first is formulating a program—an overarching set of intentions to guide the exploratory design process [38]. In this study, the program investigated how participatory visual mapping can foster an understanding of more than human temporality. Several temporal categories, including relational time, were introduced in each workshop to prob a temporal-oriented mindset among participants. The second component involves realising the program through experimental practices, turning the early intuition that initially framed the program into new experiments to reframe the problem, position the enquiry and broaden the design space [38, 46]. The third element entails synthesising findings through reflective practices [9, 38, 46]. Reflection facilitates the articulation of results and fosters theoretical advancements. Simultaneously, the experimental process is a medium for theory development and knowledge production [3]. This integrated approach enables the co-construction of theory and practice, thereby contributing to a nuanced understanding of the research inquiry.

The findings in this paper stem from three workshops that lasted 5-7 hours and were conducted as part of a larger research program in Denmark and Sweden in 2023–2024. Facilitated by colleagues and me, these workshops engaged 32 participants (Table 1). Each workshop followed a three-part structure: first, participants observed and interacted with more than human entities, developing attentiveness to more than human time. In the second phase, they created tangible depictions of more than human temporality based on their experiences. Finally, participants shared their work in a participatory visual mapping exercise on a collective canvas, fostering collaborative discussions and reflections. A detailed description of each workshop is provided.

I frame the term visual mapping following the map definition as "a diagram or other visual representation that shows the relative position of the parts of something" [40], highlighting the depictive role of the map as an artefact and articulating the map's fragmental and relational quality. The map was assembled from multiple artefacts or sketches that participants laid out in relation to each other, implying various perspectives. Its role was to support the facilitation of the discussion, the exploration of diverse interpretations, and to reveal tacit knowledge [49]. In this work, visual mapping acts as a practice of shared investigation and thinking that seeks to achieve a more sophisticated and nuanced understanding of intricacies by establishing forums for discussion through sensory experiences [57]. It allowed participants to discuss their work, relate to others' work, and identify commonalities.

3.2 Workshop description

3.2.1 Workshop 1. The first workshop was held in a hybrid format, combining in-person sessions at Malmö Harbour with online participation. It brought together 11 participants from across Europe, all researchers interested in more-than-human perspectives. Each received a digital or tangible workbook, which guided them through a series of outdoor exercises to be completed near a body of water (Table 1). These included guided meditation, close observation, writing, and sketching—activities designed to deepen sensory engagement with their surroundings. While some participants gathered at Malmö Harbour to follow the exercises collectively, others worked independently in various locations, using the workbook at their own pace. Following this initial phase of environmental attunement, participants were invited to engage with a specific more-than-human actor or assemblage through observation, sketching, photography, and quick online research (Table 1). The workshop concluded with a reflective exercise in which participants created drawings of the actor's temporal experience, materialising it through sketching, guided by three categories—time in relation to the natural world (e.g., seasons, tides, light, astronomical cycles), lifetime, and pace—alongside prompts about the nature and direction of time, and potential interruptions (Figure 1).

In the second part of the workshop, participants uploaded their sketches and visual notes to a shared Miro board (Figure 2), which served as the basis for a collective visual map. They then joined an online meeting to connect with the other participants. Facilitators first asked them to silently place their contributions on the board according to temporal categories defined in previous workshops, such as relational time, universal time, and circular time. Then, they asked participants to observe others' work, see if anything caught their attention due to similarity or contrast with their own, and mark these connections on the board. Participants were then invited to add to the temporal lexicon, adding the temporal vocabulary that emerged during their work.

In the subsequent discussion, participants presented the temporalities they had associated with the actor they followed through sketching. They shared how their experience unfolded throughout the exercise, which, in turn, prompted broader reflections on the nature of temporality. The conversation also addressed relationships between different actors, including the participants themselves. Debates emerged over the interpretation of various temporal terms, and which offered the most significant potential for fostering a

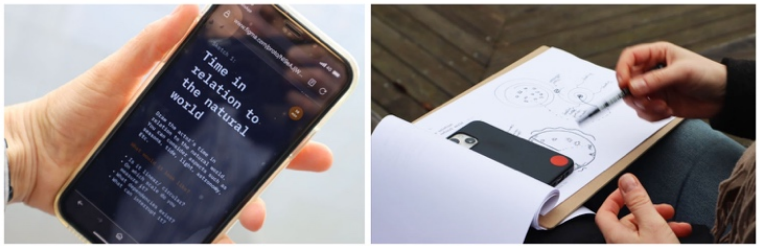


Figure 1: Workshop One. Left: The digital workbook. Right: a participant following the sketching exercises.

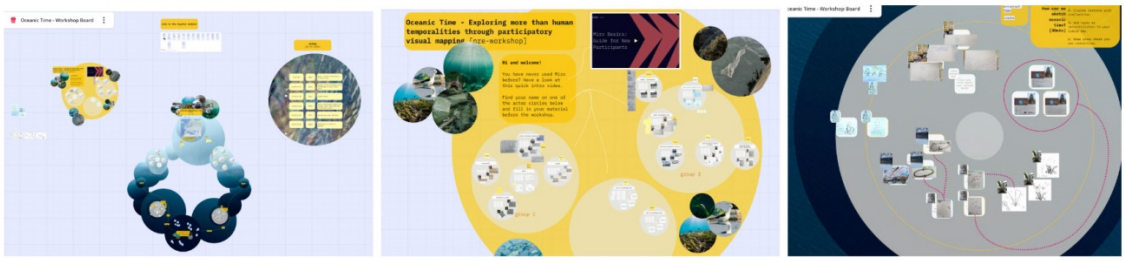


Figure 2: Workshop One. The Miro board in use. Click the link to enter the online board. https://miro.com/app/board/uXjVNz0sg4U=/



Figure 3: Workshop Two. Left: Tuning with the environment through an observation exercise. Right: an overview of some available materials and the working space.

more nuanced understanding of temporal dynamics. Finally, participants considered how the exercise might be applied or adapted within their own professional contexts.

3.2.2 Workshop 2. The second workshop adopted a similar structure and was conducted in person at the South Harbour of Copenhagen, as part of an academic course for master's students in design, involving 18 participants. Following an introductory lecture on more-than-human approaches in design, the workshop commenced with participants working in pairs to undertake the exploratory activities outlined in Table 1. In the second phase, participants were organised into groups of four based on their chosen actors. Each group worked around a shared canvas equipped with a range of materials (Figure 3) and selected a temporal category to investigate through the creation of a physical artefact. This process required participants to present and negotiate their initial research, synthesising their insights into a collective output. The activity was designed to scaffold temporal thinking in stages: beginning with observation and dialogue, advancing to sketching, and culminating in hands-on making with a variety of materials— including wood, soil, water, tape, rope, mirrors, and straws — with language serving as a continuous and integral element throughout the process.

The workshop concluded with group presentations followed by a collective discussion. Participants were asked to introduce the actors they had explored through the lens of temporality and to describe how the materialisation of temporal concepts had developed over the course of the workshop. In response, they reflected on the differences between thinking about a more-than-human actor and adopting such a perspective, highlighting the challenges this entailed. Further reflections addressed the relationships that emerged among actors, including considerations of their own positionality and the power imbalances inherent in design processes seeking to engage with more-than-human perspectives. Additional insights concerned the mapping process itself—particularly how

Table 1: An overview of the temporal probes in each workshop

Workshop no.	Exercises	Experiences	Materials used in the mapping session
1	Meditation attuned to time, observe things in the environment you associate with time, online research on lifespan and dependencies, exploring actors' time through individual sketching (incl. Probing questions).	Outdoor observation on their own time and pace, or in the harbour with other participants for two hours.	Pens and paper.
2	As above, incl. uses the materials to create an artefact that reflects the temporality of your actors - what is time for them?	A two-hour outdoor activity in pairs.	Wood, soil, water, threads, mirrors, wooden balls and blocks, tape, glue, bulbs, branches.
3	Note-taking in relation to temporal categories while experiencing the stations, based on your experience with the exhibition space and the note taken earlier, create an artefact relating to one of the temporal categories introduced before. You can build on your earlier notes and the ideas expressed by others.	Exploring an exhibition space for one hour, including the following stations: augmented moss observatory, AI-generated plant responsive to drawings of moss branches, AI-generated climate change, landscape alteration, smelling and tasting materials of various ages, co-creating an ice sculpture, listening to microbial sounds, making sand traces as you move in the space.	Wood, soil, ice, threads, mirrors, wooden balls and blocks, tape, glue, leaves, colours, feathers.



Figure 4: Workshop Three. Various stations in the exhibition space.

material choices shaped the conceptualisation of temporality and how prompting questions influenced the development of ideas.

3.2.3 Workshop 3. The third workshop was a full-day event, co-designed with fellow design researchers and conducted as part of an academic conference in Copenhagen. Held in a university classroom, it engaged 12 participants—all design researchers, including the organisers, who also took part in the activities. While the workshop's overarching aim remained consistent, the methods were adapted to the venue's context and informed by the organisers' specific expertise.

In the first phase, participants engaged with an exhibition space curated by the organising team. This space comprised a series of interactive stations where participants encountered more-than-human actors with a distinct temporal relationship (Table 1).

Participants were asked to record their reflections during the experience and post them on a shared board structured around predefined temporal categories developed in previous Workshops—such as relational time, dense time, and timehood (Figure 5). In the second phase, a canvas replicating the same temporal grid was placed on the floor. Participants were invited to review the collected reflections, construct an artefact in response, and position it within the corresponding temporal category on the canvas (Figure 5). A curated selection of materials was provided to facilitate construction.

Once the map was completed, participants shared their experiences and processes, as well as their curiosities about each other's work, and discussed the relationships that emerged among the various artefacts and ideas. The following discussion touched on several themes, including how temporal dimensions—such as scale, synchronicity, urgency, and accumulated time—can support reflection on their own design practices. Participants also considered the challenges of attuning to the world and to more-than-human

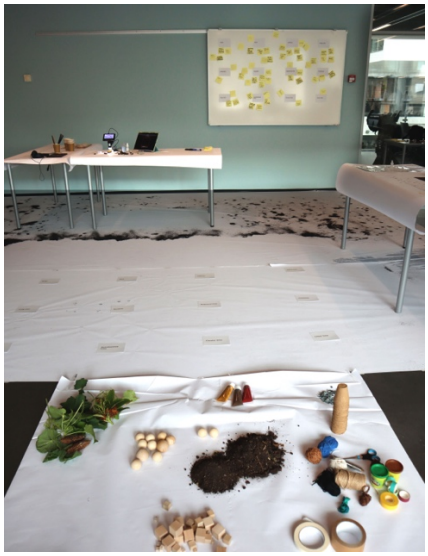


Figure 5: Workshop Three. An overview of the room, including the materials provided to the participants, the canvas with the temporal terms to place objects, the exhibition space and the wall canvas in-cluing the temporal terms and text notes.

perspectives, emphasising the need to remain aware of how knowledge emerges, how it may be mobilised across different contexts, and the role and limitations of language in this process.

Escobar et al. [20] mention that relationality is always contingent and local; therefore, it can only be expressed in concrete, material, and local iterations. This characteristic makes this type of work difficult to generalise, and, indeed, the main drawback of this research is the small sample of empirical work on which it is based.

3.3 Data collection and analysis

Across the workshops, data were collected in multiple forms: audio recordings of group and plenary discussions, photographs, participants' workbooks, facilitator notes, and content from the shared Miro board. To complement these sources, I also distributed questionnaires for participant feedback and documented informal conversations and observations through field notes and reflective summaries. As this paper focuses on visual mapping rather than the entire workshop process, the analysis draws specifically on data generated during the mapping sessions. After transcribing the conversations, I conducted a reflective thematic analysis [16], drawing on texts, notes, and images to identify patterns relevant to relational temporality. This process began with text marking, cutting and sorting key excerpts, and grouping noteworthy quotes and phrases. I then proceeded to coding, paying particular attention to repetitions, metaphors, synonyms, similarities, and contradictions [16, 17, 63]. My coding followed an abductive approach [20], which emphasises openness, curiosity, and responsiveness to the unexpected, thereby allowing opposing viewpoints to emerge as generative insights beyond the workshops' core themes. Finally, I examined participants' responses to the post-workshop questionnaire [15], which provided direct answers to targeted questions.

4 FINDINGS

In this section, I will describe four relational dimensions of temporality that emerged during the visual mapping sessions: relations between different time scales, the human capacity to perceive and synchronise with more than human time, multi-species temporal dependencies and relational workshop time.

4.1 Relations between different time scales

The coming together of various time scales was a recurring subject in the mapping sessions. A participant in Workshop Three created a model of a "relational time garden" (Figure 6) that brought together biological, geological, and human timeframes. This exercise sparked a rich discussion about the interplay between human and more than human temporal scales and how they influence and shape each other. Another participant shared her expectations, and the temporal disconnect she experienced when reflecting on her interaction with moss. She described attempting to influence the moss's state by watering it while it was already wet and wishing it would dry out again. This revealed a temporality mismatch, as she recognised that her actions were out of sync with the moss's rhythms. The relationship between temporal scales also surfaced through explorations with materials. One participant, reflecting on "duration", dipped a ball in the paint and rolled it over a surface (Figure 6). The resulting imprints made her consider the interaction dynamics: "When the ball contacts another surface, it expands its scale in some way because it activates other surfaces and materials." These insights underscore how temporal engagement with living entities or material processes can highlight the complexities and interdependencies of different timeframes. Another topic of discussion was what happens when different time scales connect momentarily. A participant suggested to call it "shared time". Another suggested "overlap time" instead, as the brief interaction is not voluntary but often occurs due to circumstances.

4.2 Multi-species temporal dependencies

In Workshop One, participants explored temporal dependencies through the observation and noticing implied in sketching. A participant who sketched the growth pace of a juniper shrub (Figure 7), imagined himself as the shrub and realised it is shaped by the actions of animals and humans interacting with it. Reflecting on interactions with seagulls, a participant highlighted the inseparability of their time from surrounding elements, such as the sea, fishermen, and boats. Observing reeds (Figure 7), another participant recognised their dependence on cows grazing on them and humans cutting them down in autumn.

A participant who explored Goos in Malmo Harbour (Figure 8) reflected on how their migration temporality is affected by climate change, and how their behaviour and health are affected by human interventions. She posed some critical questions regarding the connection to the ecosystem, relating to "polyphonic patterns" created between temporal circles of various actors.

The rich selection of materials allowed participants to develop their thinking about temporal relationships to a more abstract level.



Figure 6: Left: The relational time garden. Right: a ball deepens in the paint.

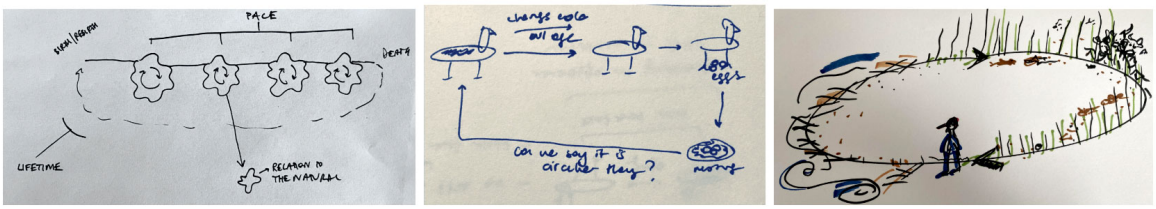


Figure 7: Left: a sketch of a Juniper Shrub cycle. Centre: a sketch of Seagull's lifecycle. Right: an annual cycle of a reed.

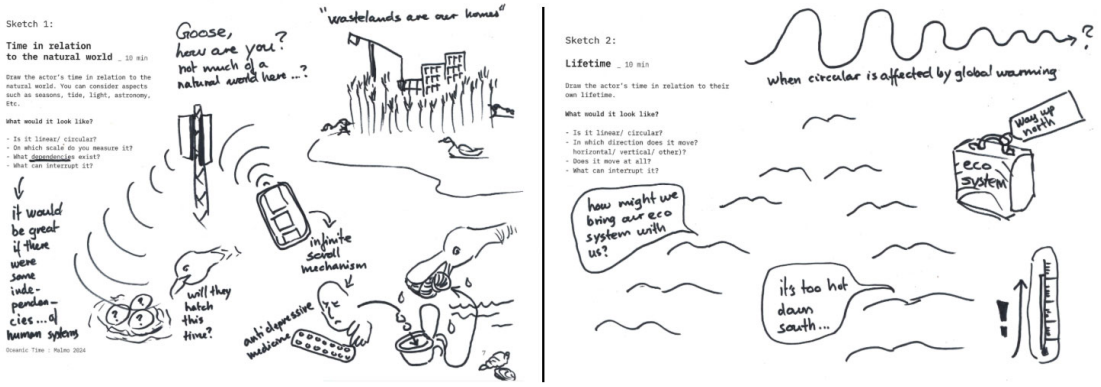


Figure 8: An illustration of Goos dependencies.

After rolling a paint-dipped ball across the canvas (Figure 9), one participant reflected on the complex factors influencing its movement: "My gesture set it off, but the air pressure, paint thickness, and the paper's texture all played a role in its path. It's not just my action—the environment shapes it." Another participant extended this idea, noting how even a tiny gesture can trigger ongoing ripples: "The ball now lies there, but later, someone will clean it, walk in a direction, and something else will happen. The tiny gesture set off a chain of events that circulated." This highlighted the intricate web of temporal and material interactions.

Four participants who worked together in Workshop Two explored the temporality of two different Weeping Willow trees with which they interacted as couples (Figure 9). When synthesising their experience into a single artefact, they highlighted their observed differing growth conditions. While the first tree thrived in a central park, nurtured by human care, the other emerged from a pavement crack in a neglected area. Participants represented the trees metaphorically as a ball game with two courts: one smooth and streamlined, enabling fast growth, and the other filled with obstacles, slowing progress. A participant noted, "These are trees in a garden controlled by humans, so we depicted it linearly. It would look different if it were in a forest." This depiction illustrates how the pace and growth of the trees reflect their relationship with their respective environments.

In the Third Workshop, soil was explored as a radical case of merging temporalities, where individual actors and species become indistinguishable from one another. A participant created an "arch of accumulated time" (Figure 10) to illustrate biodegradability and composting. His composition ranged from "recently fallen materials" to those that had "been lying for a while," mixing fragments

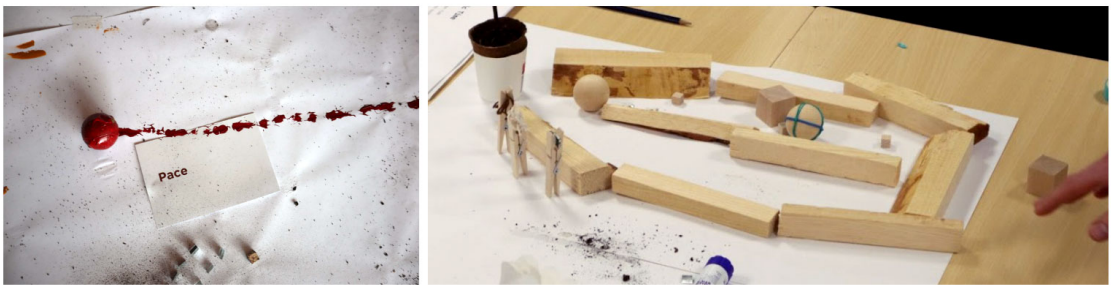


Figure 9: Left: A rolling ball. Right: A temporality of two Weeping Willow trees.



Figure 10: Left: The arch of accumulated time. Right: Composting, compost time, and soil time.

and placing a melting ice cube on top to transform the materials. Another participant who worked with soil suggested that "accumulated participation" is made by microbes, earthworms, bacteria, water, nitrogen, and carbon, which accumulate over time to create the soil, which is "always in this process of accumulated time." The participant argued that this accumulation exists even in an amount as small as a quarter of a teaspoon of soil, which he chose to use (Figure 10). In both works, the soil was approached as a living entity that was generated and changed over time through the participation of different entities and their temporal relationships.

4.3 The human capacity to perceive and synchronise with more than human time

The human ability to perceive and synchronise with more than human temporalities was a central theme that emerged from participants' work and during the discussions. One participant who engaged with the exhibition space referred to the concept of 'dense time' by stacking various materials around a plastic container (Figure 11), highlighting this challenge through a small artefact created after engaging with the exhibition space. She explained: "My ability to be with so many temporalities is limited. I have a certain capacity within the workshop time, attention span, and my temporality. So, I mostly think about how much I can take in, what I can get out of it, and how long it will take. I feel like I am somehow stuck inside this bottle in the middle of all that variety and cannot reach it" (Figure 11). Another participant resonated with this sentiment, emphasising: "Our attention spans, our attention frame, and the degree to which we can tune our attention toward the notion of the world are a serious ongoing challenge." The question of accessing other-than-human temporalities was also explored by another participant who created a communication-inspired tool resembling two ends connected by a string (Figure 11). The instrument allowed her to listen and develop a relationship with something she perceived as inaccessible. This sparked a discussion about the tool's bidirectionality. One participant inquired about how the tool might function from a more than human perspective, highlighting the asymmetry of human-centric experiences. Amplification, she responded, might help bridge the gap, saying, "Amplifying something makes it noticeable for me," underlining how tools and attention shape our capacity to engage with more than human time.

In response to each other's work, Participants ask: How do we pay attention to others? Do we integrate what we observe into our lives and become aware of the cycles that surround us? Do our relationships with them shape this awareness, and if so, why do we engage in this process? When multiple dynamics unfold simultaneously? And can we somehow sense its collective energy? Participants also observed that it is easier to recognise more than human temporalities that we share, such as cyclical time. To address the limited understanding of time beyond human comprehension, a participant suggested recruiting an environmental scientist or biologist to provide participants with expert knowledge. Another participant indicated that drawing could draw our awareness to



Figure 11: Left: The capacity to absorb more than human temporalities. Right: A communication device.

the perspective from which we speak. Are we outside observers, or can we experience time as if we were actors ourselves?

4.4 Creating shared time

The workshops offered participants only a brief window — spanning just a few hours — to engage with more-than-human actors. Constrained by academic class schedules and conference schedules, participants' availability, and the cold weather, these interactions often took the form of fleeting, momentary engagements or 'temporal snapshots'. However, several elements extended the workshops' temporality beyond these immediate encounters. In Workshop One, remote participants followed the exercises at their own pace, often revisiting familiar environments they had experienced across different seasons and times of day. One participant, for instance, explored a beach where he had grown up, drawing on decades of personal memory. In this way, their observations unfolded along an expanded temporal axis. In Workshops Two and Three, many of the organic materials provided were subject to change over time, introducing a built-in temporal dimension. Additional layers of time were introduced through the accumulation of online information and through observation enabled by focused attention [74]. Relational time emerged most vividly through what participants described as "shared time"—the moments spent together during the workshops, marked by collective listening, co-creation, and the development of shared thought. These discussions gave rise to new temporal terminology — such as accumulated time, time overlap, collaborative attentions, and the clash of temporalities — that served as poetic and metaphorical ideas rather than consolidating temporal terminology, and that reflected an evolving and nuanced understanding of time. One participant in the first workshop proposed the term timehood to capture the richness of these layered experiences. Another, in the third workshop, described the output of visual mapping as "a memory castle of concepts and perspectives."

5 DISCUSSION

As the workshops unfolded, it became clear that participants' understanding of time was shaped not only by what they observed, but also by what they could observe—what was available to their senses, attention, and modes of engagement. This raises a deeper epistemological and ethical question at the heart of relational temporality: what counts? [7, 11, 42, 58]. This question asks who or what we choose to attend to, recognise, or value [55]; what we frame as a disturbance [70]; who is seen as an actor [42]; and what forms of significance emerge within a community or situated context [7]. The workshop findings suggest that these judgments are not neutral or purely cognitive—they are shaped by embodied practices, sensory access, and material engagements [49]. For instance, what participants noticed in sketching or tactile exercises, the time scales they could relate to, and the dependencies they recognised were conditioned by their proximity to nonhuman actors, their sensory thresholds, their cultural frames of reference, and their conversations with peers. This supports arguments by scholars who have shown that temporality is not a fixed abstraction but emerges through relational and material practices. It also echoes Law and Joks' [44] view of how forces—human and more-than-human—act on and through us to shape temporal experience.

In response to these constraints on perception and recognition, I propose the concept of temporal relational sensitivity as a necessary design capability. It extends the idea of relational sensitivity [2, 31, 32] into the temporal dimension, urging designers and participants to reflect not only on who they are entangled with but also how time is experienced, valued, and mediated across these relationships. Cultivating temporal relational sensitivity involves attuning to the plurality of temporalities, whether the slow recovery of degraded land, the rapid cycles of microbial life, or the imperceptible rhythms of infrastructures and climate systems. This kind of sensitivity challenges normative, extractivist orientations toward time—those that prioritise immediacy, efficiency, and predictability—and invites a shift toward attentiveness, care, and long-term responsibility. Participatory visual mapping, as this study demonstrates, offers a concrete and generative method for nurturing this capability by making visible what often escapes attention and by fostering dialogic, multi-perspectival encounters with time.

The findings demonstrate that visual mapping facilitates the development of a relational perspective by revealing multispecies temporal dependencies, highlighting diverse timescales, challenging the human capacity to perceive and synchronise with more-than-human time, and creating shared time. These insights point to the broader relevance of temporal relational sensitivity as both a

Table 2: A summary of the three practical steps of participatory visual mapping, as suggested by this work

Step	Format	Purpose	Methods in use	Facilitators' role
Attunement	Individually or in small groups	An invitation to slow down, observe and engage sensorially with the surroundings	Writing, sketching, photography, online research, meditation, observation, sensory engagement.	Guide participants through the sequence of activities and use prompting questions to draw their attention to the relational aspects of time (e.g., who are you looking at? How long does it live? On which scale do you measure its time? What dependencies exist? What can interrupt it?)
Visualisation and materialisation	Individually or in small groups	To give shape to ideas and mediate reflections. Help participants to develop and externalise temporal insights.	Sketching, drawing, modelling, building, playing with materials.	Making the material's properties visible (e.g., how organic materials change over time) and encouraging reflection on meaning-making during the process (e.g., asking whether the artefact reflects an initial understanding of time or whether that understanding developed through the act of making).
Mapping	Collectively	A situated and facilitated practice of collective sense-making, sharing, creating and negotiating temporal or relational insights.	Online, using a digital canvas) or in person, using a physical canvas on a table, whiteboard or on the floor.	Steer the conversation toward an understanding of relational aspects of time by surfacing insights (e.g., whose time is involved, how time emerges between different actors, and how the workshop's temporality shapes the outcomes), articulating tensions, and reinforcing emerging terms.

practical design orientation for complex, interdependent systems and a theoretical lens.

From a practical perspective, the method's visual qualities helped make the unseen visible, drawing attention to temporal dimensions often excluded from immediate awareness. Enacting participatory visual mapping requires cultivating conditions where participants can think with materials, environments, and more-than-human actors rather than merely about them. The process begins with attunement—inviting participants to slow down, observe, and engage sensorially with their surroundings. This may take the form of guided walks, situated exercises, or open-ended prompts that foreground embodied and ecological awareness. Visualisation and materialisation are then used not as tools of representation but as mediators of reflection: sketching, arranging, or building with diverse materials helps participants develop, externalise and negotiate temporal or relational insights. Facilitators play a key role in holding the process open—balancing structure and emergence, guiding without prescribing outcomes, and ensuring that diverse voices and temporalities can coexist on the same map. Reflection and dialogue are integral throughout, enabling participants to trace how their own positionalities and assumptions shape what becomes visible. Ultimately, participatory visual mapping is less a fixed method than a situated practice of collective sensemaking, one that invites ongoing experimentation with how knowledge, relations, and temporalities are rendered and reconfigured through design. At the same time, this situated character introduces a productive tension: the practice requires careful guidance and framing yet resists standardisation. Its methodological value lies not in reproducibility but in responsiveness to context, participants, and relations as they unfold. Making this tension explicit foregrounds the core dynamic of this contribution—offering direction while refusing universal prescription and treating situatedness not as a limitation but as its primary generative force.

Theoretically, the workshop not only surfaced knowledge about relational temporality but also enacted it in practice. Participants shared reflections—referred to by one as a "memory castle"—enabled the co-creation of new language, metaphors, and insights. Terms like "shared time," "common time," and "timehood" emerged as participants built on one another's contributions, posed provocative questions, and drew on personal memories. These dialogic and imaginative exchanges illustrate how participatory visual mapping can function not only as a tool for documentation or ideation but also as a relational and temporal medium. This challenges conventional design paradigms that centre on individual, human, and

Table 3: A summary of key synergies and differences between the technique proposed in this paper and other work exploring relational temporality in design

Category	References	Main synergies	The novelty of this paper in relation to the work
Temporalities in participatory design processes, projects and events	[4, 23, 34, 54, 64, 65]	Temporality as a result of the involvement and interaction between multiple stakeholders and circumstances.	A more-than-human perspective.
The coming together of more-than-human temporalities through the lens of timekeeping.	[33]	Recognising the existence of temporalities beyond the human and the challenge to perceive them.	A relational understanding of temporality and a different methodology.
Time-oriented visual mapping in participatory design.	[75]	Visual mapping as a method for unpacking temporal complexity	More-than-human perspective.
Relational understanding of more-than-human temporality.	[11, 34, 50, 53]	Relational understanding of more-than-human temporality.	Suggesting a concrete methodology to address the theoretical challenges.
Using a solution-oriented visual approach to address concrete ecological challenges.	[57]	Using a visual method to address more-than-human temporal challenges.	A different epistemic approach, unpacking temporality through the lens of relationality.

efficiency-based understandings of time. Instead, the method introduces a model of temporal relational sensitivity, inviting designers to engage with time as co-constituted through multispecies interactions, material affordances, and collective meaning-making. In doing so, it extends relational approaches in participatory design by foregrounding temporality as a central, rather than peripheral, concern, particularly in relation to more-than-human and socio-ecological systems.

This paper offers two main contributions to participatory design. First, it broadens the conceptualisation of relational temporalities by moving beyond human-centred framings and drawing on theoretical insights from adjacent fields. Second, it demonstrates how this expanded understanding can be enacted in practice through the methodology of participatory visual mapping, whose collaborative and multisensory nature challenges routine, individualised, and extractivist orientations toward time in design processes. Together, these contributions advance an orientation of temporal relational sensitivity—one that invites designers to attune to the multiplicity of temporal relations shaping more-than-human worlds. In doing so, the paper extends participatory design's ethical and ecological scope, opening pathways for practices that respond to time not as a background condition, but as a shared and co-created field of relation. Table 2 summarises these contributions in relation to existing design research on temporality.

Acknowledgments

I would like to thank my collaborators in designing and facilitating the workshops, as well as all participants, for their creative engagement and for generously sharing their thoughts, experiences, and insights. I also thank the reviewers for their careful reading and thoughtful, thorough comments, which significantly strengthened this paper.

References

[1] David Abram. 2017. The spell of the sensuous: perception and language in a more-than-human world. Vintage Books, New York.
[2] Yoko Akama and Ann Light. 2018. Practices of readiness: punctuation, poise and the contingencies of participatory design. In Proceedings of the 15th Participatory Design Conference: Full Papers - Volume 1, August 20, 2018. ACM, Hasselt and Genk Belgium, 1–12. https://doi.org/10.1145/3210586.3210594
[3] Yoko Akama, Ann Light, and Takahito Kamihira. 2020. Expanding Participation to Design with More-Than-Human Concerns. In Proceedings of the 16th Participatory Design Conference 2020 - Participation(s) Otherwise - Volume 1 (PDC '20), June 18, 2020. Association for Computing Machinery, New York, NY, USA, 1–11. https://doi.org/10.1145/3385010.3385016
[4] Andrea Armstrong, Emma Flynn, Karen Salt, Jo Briggs, Rachel Clarke, John Vines, and Alistair MacDonald. 2023. Trust and temporality in participatory research. Qual. Res. 23, 4 (August 2023), 1000–1021. https://doi.org/10.1177/14687941211065163
[5] Anne Louise Bang and Mette Agger Eriksen. 2019. Experiments all the way in programmatic design research. Artifact 6, 1–2 (December 2019), 8.1-8.20. https://doi.org/10.1386/art_00008_1
[6] Pratima (Tima) Bansal, Juliane Reinecke, Roy Suddaby, and Ann Langley. 2022. Temporal Work: The Strategic Organization of Time. Strategy. Organ. 20, 1 (February 2022), 6–19. https://doi.org/10.1177/14761270221081332
[7] Michelle Bastian. 2012. Fatally Confused: Telling the Time in the Midst of Ecological Crises. Environ. Philos. 9, 1 (2012), 23–48. https://doi.org/10.5840/envirophil2012913
[8] Michelle Bastian. 2023. Is long-term thinking a trap? Chronowashing, temporal narcissism and the time machines of racism. Environ. Humanities. 16, 2 (October 2023). Retrieved December 5, 2023 from https://www.research.ed.ac.uk/en/publications/is-long-term-thinking-a-trap-chronowashing-temporal-narcissism-an
[9] Michelle Bastian and Rowan Bayliss Hawitt. 2023. Multi-species, ecological and climate change temporalities: Opening a dialogue with phenology. Environ. Plan. E Nat. Space 6, 2 (June 2023), 1074–1097. https://doi.org/10.1177/25148486221111784
[10] Bernadette Bensaude-Vincent. 2022. Rethinking time in response to the Anthropocene: From timescales to timescapes. Anthr. Rev. 9, 2 (August 2022), 206–219. https://doi.org/10.1177/20530196211006888
[11] Riel Bessai, Roy Bendor, and Ruud Balkenende. 2024. Designing with more-than-human temporalities. In Proceedings of DRS, June 23, 2024. Design Research Society. https://doi.org/10.21606/drs.2024.438
[12] Thomas Binder and Johan Redström. 2006. Exemplary Design Research. In Wonderground, 2006. Lisbon, Portugal. Retrieved from https://dl.designresearchsociety.org/drs-conference-papers/drs2006/researchpapers/51
[13] Marcia Bjornerud. 2018. Timefulness: how thinking like a geologist can help save the world. Princeton University Press, Princeton.

[14] Maylei Blackwell. 2023. Indian Time: Temporality, Gender, Mobility. Mex. Stud. Mex. 39, 1 (February 2023), 92–116. https://doi.org/10.1525/msem.2023.39.1.92

[15] Norman Blaikie and Jan Priest. 2018. Designing Social Research: The Logic of Anticipation. Polity, Medford, MA.

[16] Virginia Braun and Victoria Clarke. 2022. Thematic analysis: a practical guide. SAGE, Los Angeles London New Delhi Singapore Washington DC Melbourne.

[17] Svend Brinkmann. 2014. Doing Without Data. Qual. Inq. 20, 6 (July 2014), 720–725. https://doi.org/10.1177/1077800414530254

[18] Jacob Buur, Bernd Ankenbrand, and Robb Mitchell. 2013. Participatory business modelling. CoDesign 9, 1 (March 2013), 55–71. https://doi.org/10.1080/15710882.2012.760609

[19] Dipesh Chakrabarty. 2018. Anthropocene Time. Hist. Theory 57, 1 (2018), 5–32. https://doi.org/10.1111/hith.12044

[20] Kathy Charmaz and Linda Liska Belgrave. 2019. Thinking About Data With Grounded Theory. Qual. Inq. 25, 8 (October 2019), 743–753. https://doi.org/10.1177/1077800418809455

[21] David Couzens Hoy. 2009. The Time of Our Lives: A Critical History of Temporality. MIT Press, Cambridge, MA. Retrieved October 19, 2022 from https://doi.org/10.1177/0961463X10394958

[22] María Puig De La Bellacasa. 2017. Matters of Care: Speculative Ethics in More than Human Worlds. University of Minnesota Press. Retrieved June 11, 2024 from https://www.jstor.org/stable/10.5749/j.ctt1mmfspt

[23] Chiara Del Gaudio, Carlo Franzato, and Alfredo Jefferson de Oliveira. 2017. The challenge of time in community-based participatory design. URBAN Des. Int. 22, 2 (May 2017), 113–126. https://doi.org/10.1057/s41289-016-0017-5

[24] Martin Dodge, Rob Kitchin, and Chris R. Perkins (Eds.). 2011. Rethinking maps: new frontiers in cartographic theory (Paperback ed ed.). Routledge, London.

[25] Mads Ejsing, Lars Tønder, Ingrid Helene Brandt Jensen, and Janus Hansen. 2024. Do we have time for democracy? Climate action and the problem of time in the Anthropocene. Anthr. Rev. (September 2024), 20530196241279564. https://doi.org/10.1177/20530196241279564

[26] Arturo Escobar, Michal Osterweil, and Kriti Sharma. 2024. Relationality: an emergent politics of life beyond the human. Bloomsbury Visual Arts, Bloomsbury Publishing Plc, London, UK; New York, NY.

[27] Ignacio Farías. 2017. An Idiotic Catalyst: Accelerating the Slowing Down of Thinking and Action. Cult. Anthropol. 32, 1 (February 2017), 35–41. https://doi.org/10.14506/ca32.1.05

[28] Tony Fry (Ed.). 1999. A new design philosophy: an introduction to defuturing. UNSW Press, Sydney.

[29] Elaine Gan. 2017. Timing rice: an inquiry into more-than-human temporalities of the Anthropocene Elaine Gan. New Form. 92, 92 (September 2017), 87–101. https://doi.org/10.3898/newf:92.06.2017

[30] Katie Gaudion, Ashley Hall, Jeremy Myerson, and Liz Pellicano. 2015. A designer's approach: how can autistic adults with learning disabilities be involved in the design process? CoDesign 11, 1 (January 2015), 49–69. https://doi.org/10.1080/15710882.2014.997829

[31] Laura Gottlieb. 2022. Relational sensitivity and participatory practice: Exploring poise and punctuation through an empirical study. In Proceedings of the Participatory Design Conference 2022 - Volume 2, August 19, 2022. ACM, Newcastle upon Tyne United Kingdom, 186–190. https://doi.org/10.1145/3537797.3537872

[32] Laura Gottlieb. 2024. Relational sensitivity in participatory design. CoDesign 20, 3 (July 2024), 440–460. https://doi.org/10.1080/15710882.2024.2317943

[33] Rupert Griffiths. 2023. Time and the Anthropocene: Making more-than-human temporalities legible through environmental observations and creative methods. Time Soc. 32, 4 (November 2023), 461–487. https://doi.org/10.1177/0961463X231202928

[34] Niall Hayes, Lucas D. Introna, and Noel Cass. 2021. Participatory Design as the Temporal Flow of Coalescing Participatory Lines. Comput. Support. Coop. Work CSCW 30, 4 (August 2021), 507–538. https://doi.org/10.1007/s10606-021-09405-4

[35] Liesbeth Huybrechts, Rachel Charlotte Smith, Heike Winschiers-Theophilus, Daria Loi, and Jesper Simonsen. 2024. Participatory Design in a Changing World: A Situated Action. In Proceedings of the Participatory Design Conference 2024: Situated Actions, Doctoral Colloquium, PDC places, Communities - Volume 3, August 11, 2024. ACM, Sibu Malaysia, 59–61. https://doi.org/10.1145/3661456.3666065

[36] Yuta Ikeya and Bahareh Barati. 2023. Designing an Algal Relay Computer: A Critical Orientation in Exploring More-than-Human Temporality. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, April 19, 2023. ACM, Hamburg Germany, 1–6. https://doi.org/10.1145/3544549.3585874

[37] Tim Ingold. 1993. The Temporality of the Landscape. World Archaeol. 25, 2 (1993), 152–174.

[38] Solveig Joks and John P. Dunne. 2017. Indigeneity, science and difference: notes on the politics of how. October 11, 2017.

[39] Asnath Paula Kambunga, Rachel Charlotte Smith, Heike Winschiers-Theophilus, and Ton Otto. 2023. Decolonial design practices: Creating safe spaces for plural voices on contested pasts, presents, and futures. Des. Stud. 86, (May 2023), 101170. https://doi.org/10.1016/j.destud.2023.101170

[40] Helena Karasti, Karen S. Baker, and Florence Millerand. 2010. Infrastructure Time: Long-term Matters in Collaborative Development. Comput. Support. Coop. Work CSCW 19, 3–4 (August 2010), 377–415. https://doi.org/10.1007/s10606-010-9113-z

[41] Rob Kitchin and Martin Dodge. 2007. Rethinking maps. Prog. Hum. Geogr. 31, 3 (June 2007), 331–344. https://doi.org/10.1177/0309132507077082

[42] Bruno Latour. 1997. Trains of Thought The Fifth Dimension of Time and its Fabrication. Common Knowl. Vol.6 n°3, (1997), 170–191.

[43] Thomas Laurien, Li Jönsson, Petra Lilja, Kristina Lindström, Erik Sandelin, and Åsa Ståhl. 2022. An Emerging Posthumanist Design Landscape. In Palgrave Handbook of Critical Posthumanism, Stefan Herbrechter, Ivan Callus, Manuela Rossini, Marija Grech, Megen de Bruin-Molé and Christopher John Müller (eds.). Springer International Publishing, Cham, 469–491. https://doi.org/10.1007/978-3-031-04958-3_42

[44] John Law and Solveig Joks. 2019. Indigeneity, Science, and Difference: Notes on the Politics of How. Sci. Technol. Hum. Values 44, 3 (May 2019), 424–447. https://doi.org/10.1177/0162243918793942

[45] Ann Light. 2015. Troubling Futures: Can Participatory Design Research provide a Constitutive Anthropology for the 21st Century? Interact. Des. Archit. 26 (September 2015), 81–94. https://doi.org/10.55612/s-5002-026-005

[46] Ann Light. 2024. More-than-Human Participatory Approaches for Design: Method and Function in Making Relations. In Participatory Design Conference 2024, August 11, 2024. ACM, Sibu Malaysia, 1–6. https://doi.org/10.1145/3661455.3669862

[47] Jonas Löwgren, Henrik Svarrer Larsen, and Mads Hobye. 2013. Towards programmatic design research. Des. Learn. 6, 1–2 (December 2013), 80. https://doi.org/10.2478/dfl-2014-0017

[48] Machado De Oliveira Vanessa. 2021. Hospicing Modernity: Facing humamity's wrongs and the implications for social activism. North Atlantic Books, Berkeley, California.

[49] Maarit Mäkelä. 2019. A Nourishing Dialogue with the Material Environment by Maarit Mäkelä. In Design and Nature A Partnership (First Edition). Routledge.

[50] Edward Masui and Hazuki Kosaka. 2024. One Thousand Years of Infrastructuring Katsuobushi: Aligning Temporalities within More-Than-Human Entanglements. In Participatory Design Conference 2024, August 11, 2024. ACM, Sibu Malaysia, 147–152. https://doi.org/10.1145/3661455.3669883

[51] Merriam-Webster dictionary. 2024. Definition of MAP. Retrieved May 24, 2024 from https://www.merriam-webster.com/dictionary/map

[52] Giordano Nanni. 2013. The colonisation of time: ritual, routine and resistance in the British Empire (Paperback edition ed.). Manchester University Press, Manchester; New York.

[53] Gizem Oktay, Minha Lee, Bahareh Barati, and Ron Wakkary. 2024. Caring through: Engaging with temporality of care in more-than-human design. In Proceedings of the Halfway to the Future Symposium, October 21, 2024. ACM, Santa Cruz CA USA, 1–8. https://doi.org/10.1145/3686169.3686211

[54] Doenja Oogjes and Audrey Desjardins. 2024. A temporal vocabulary of Design Events for Research through Design. In Proceedings of the CHI Conference on Human Factors in Computing Systems, May 11, 2024. ACM, Honolulu HI USA, 1–12. https://doi.org/10.1145/3613904.3642560

[55] Suvi Pihkala and Helena Karasti. 2022. Towards Response-able PD: Putting Feminist New Materialisms to Work in the Practices of Participatory Design. In Participatory Design Conference 2022: Volume 1, August 19, 2022. ACM, Newcastle upon Tyne United Kingdom, 98–108. https://doi.org/10.1145/3536169.3537784

[56] Larissa Pschetz and Michelle Bastian. 2018. Temporal Design: Rethinking time in design. Des. Stud. 56, (May 2018), 169–184. https://doi.org/10.1016/j.destud.2017.10.007

[57] Larissa Pschetz, Sussane Wieland, Michelle Bastian, and Maike Gebker. 2024. Designing Temporal Ecologies: Reframing Multispecies Temporalities Through Design. June 23, 2024. . https://doi.org/10.21606/drs.2024.1068

[58] María Puig de la Bellacasa. 2017. Matters of care: speculative ethics in more than human worlds. University of Minnesota Press, Minneapolis.

[59] Amon Rapp. 2022. How do people experience the temporality of everyday life changes? Towards the exploration of existential time in HCI. Int. J. Hum.-Comput. Stud. 167, (November 2022), 102899. https://doi.org/10.1016/j.ijhcs.2022.102899

[60] Johan Redström. 2021. Research through and through design. Artifact 8, 1 (December 2021), 16.1-16.19. https://doi.org/10.1386/art_00016_1

[61] Mark Rifkin. 2017. Beyond Settler Time: Temporal Sovereignty and Indigenous Self-Determination. Duke University Press, Durham. https://doi.org/10.1515/9780822373421

[62] Micol Rispoli and Tomás Criado. 2024. Design Before Design: Learning to be Affected by Neurodiverse Spatial Practices. Des. Cult. 16, 3 (September 2024), 357–381. https://doi.org/10.1080/17547075.2024.2375453

[63] Gery W. Ryan and H. Russell Bernard. 2003. Techniques to Identify Themes. Field Methods 15, 1 (February 2003), 85–109. https://doi.org/10.1177/1525822X02239569

[64] Joanna Saad-Sulonen, Eva Eriksson, Kim Halskov, Helena Karasti, and John Vines. 2018. Unfolding participation over time: temporal lenses in participatory design. CoDesign 14, 1 (January 2018), 4–16. https://doi.org/10.1080/15710882.2018.1426773

[65] Joanna Saad-Sulonen and Thomas and Watkin. 2025. Reframing temporalities of social design: a duoethnographic reflection in hindsight. CoDesign 0, 0 (2025), 1–17. https://doi.org/10.1080/15710882.2025.2476626
[66] Elizabeth B.-N. Sanders and Pieter Jan Stappers. 2012. Convivial design toolbox: generative research for the front end of design. BIS, Amsterdam.
[67] Herbert A. Simon. 1969. The sciences of the artificial. MIT Press, Cambridge, MA, US.
[68] Manuela Taboada, Jane Turner, Jen Seevinck, and Marcus Foth. 2024. A world-building approach for pluriversality in codesign. CoDesign 0, 0 (July 2024), 1–24. https://doi.org/10.1080/15710882.2024.2349607
[69] Juan Telleria. 2024. Temporality in the United Nations 2030 Agenda: development or rupture? Futures (November 2024), 103498. https://doi.org/10.1016/j.futures.2024.103498
[70] Anna Lowenhaupt Tsing. 2015. The mushroom at the end of the world: on the possibility of life in capitalist ruins (New paperback printing ed.). Princeton University Press, Princeton Oxford.
[71] Kristof Vrancken. 2018. mijnKOOL, organic photography as a participatory design method. In Proceedings of the 15th Participatory Design Conference: Short Papers, Situated Actions, Workshops and Tutorial - Volume 2, August 20, 2018. ACM, Hasselt and Genk Belgium, 1–3. https://doi.org/10.1145/3210604.3214364
[72] Catherine Wieczorek and Laura Forlano. 2024. Reimagining Temporality: Exploring the Intersection of Time and Trauma in Design Research. DRS Bienn. Conf. Ser. (June 2024). Retrieved from https://dl.designresearchsociety.org/drs-conference-papers/drs2024/researchpapers/321
[73] Nina Williams. 2016. Creative processes: From interventions in art to intervallic experiments through Bergson. Environ. Plan. Econ. Space 48, 8 (August 2016), 1549–1564. https://doi.org/10.1177/0308518X16642769
[74] Hadas Zohar, Luca Simeone, Amalia de Götzen, and Nicola Morelli. 2024. When we talk about time, we mean many different things: employing visual mapping to think through more-than-human temporalities in participatory design. Front. Commun. 9, (August 2024). https://doi.org/10.3389/fcomm.2024.1440257
[75] Mela Zuljevic, Barbara Roosen, and Liesbeth Huybrechts. 2023. Thinging with the past: co-designing a slow road network by mediating between the historical landscape and the design space. CoDesign 19, 3 (July 2023), 194–213. https://doi.org/10.1080/15710882.2022.2038633